IBM COGNOS ANALYTICS

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Project Title: SpicePot Restaurant Chain – Sales & Customer Analytics

IBM Cognos Analytics – Case Study

SpicePot Restaurant Chain – Sales & Customer Analytics

Real-Time Business Scenario

SpicePot is a multi-cuisine restaurant chain with 25 outlets across South India offering dine-in, Takeaway, and home delivery. The operations head has observed that weekend peak hours are causing Long waiting times, certain menu items are consistently underperforming, and delivery orders are Increasing customer complaints due to cold food and long delivery times. IBM Cognos Analytics is Being used to analyze sales, menu performance, and service quality.

Case Study Questions

Q1[Reporting]

Create a Menu Performance Report showing Menu Item, Category, Total Quantity Ordered, Total Revenue, and Contribution % to Total Revenue. Apply Conditional Styling: Items Contributing < 1% to revenue in Red (consider removing). Add prompts for Outlet City and Category. Include Header with SpicePot branding, Footer with page number. Use Palatino Linotype font , size 11.

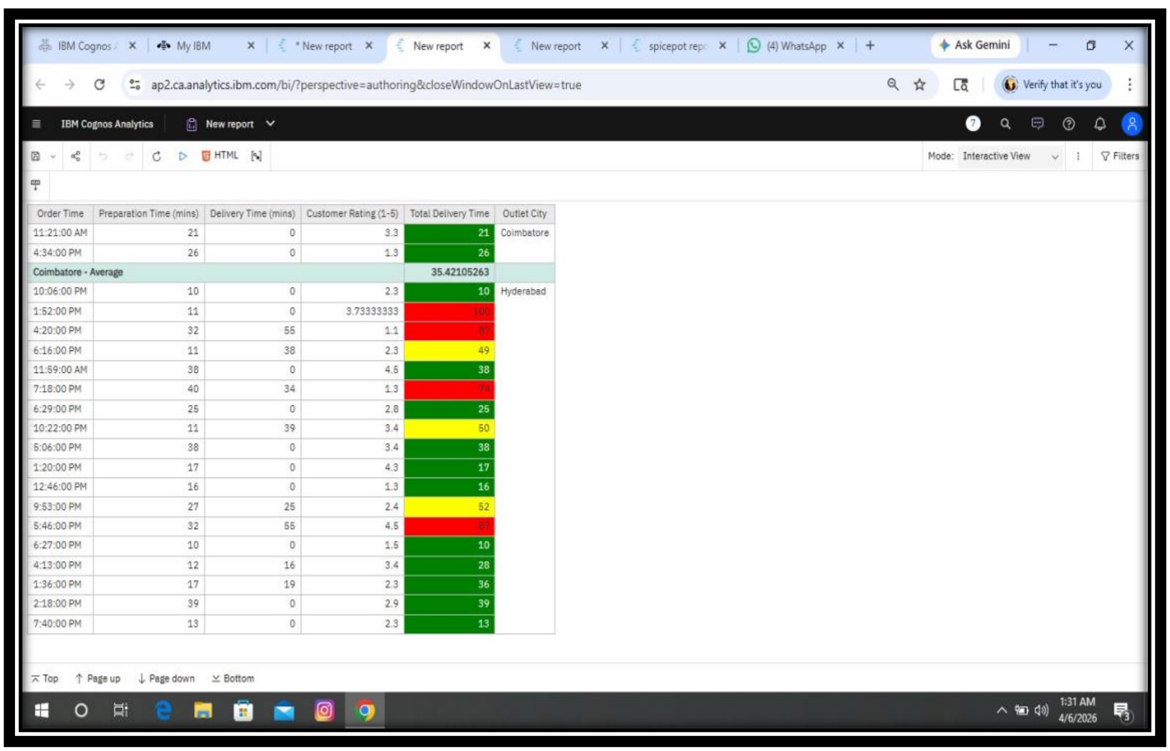
Menu Item	Category	Quantity Ordered	Total Bill	Contribution %
Brownie	Desserts	21	10885.54	100.00%
Cold Coffee	Beverages	33	16876.29	100.00%
Fish Fry	Starters	32	13542.79	100.00%
Chicken 65	Starters	19	6549.65	100.00%
Butter Chicken	Mains	23	12442.38	100.00%
Gobi Manchurian	Starters	19	8882.6	100.00%
Biryani	Mains	26	9992.2	100.00%
Paneer Tikka	Starters	23	8043.05	100.00%
Paneer Butter Masala	Mains	21	6652.45	100.00%
Chai	Beverages	37	14904.47	100.00%
Ice Cream	Desserts	50	21108.84	100.00%
Grilled Fish	Mains	39	14489.01	100.00%
Halwa	Desserts	22	9306.43	100.00%
Soft Drink	Beverages	37	7548.46	100.00%
Dal Makhani	Mains	40	18437.56	100.00%
Gulab Jamun	Desserts	34	13834.28	100.00%
Kheer	Desserts	20	10167.96	100.00%
Fresh Juice	Beverages	35	14704.87	100.00%
Veg Spring Roll	Starters	26	13758.43	100.00%
Lassi	Beverages	19	10290.16	100.00%

Menu Performance Report(Explanation):

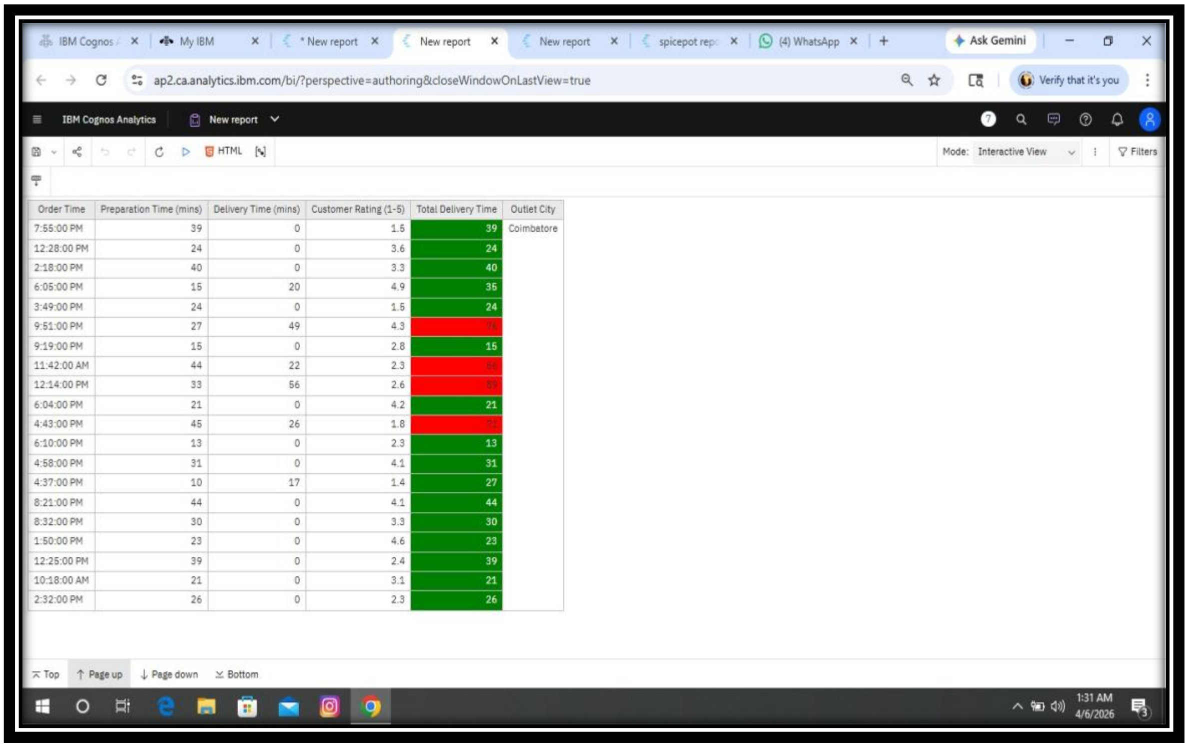
In IBM Cognos Analytics, a Menu Performance Report is created to analyze the performance of different menu items in the SpicePot restaurant chain. The report includes Menu Item, Category, Total Quantity Ordered, Total Revenue, and Contribution percentage to total revenue. Conditional formatting is applied to highlight items with less than 1% contribution in red color to identify low-performing items. Prompts are added for Outlet City and Category to filter the data easily. The report also includes a header with SpicePot branding and a footer with page numbers. The font style used is Palatino Linotype with size 11 for better readability.

Q2 [REPORTING]

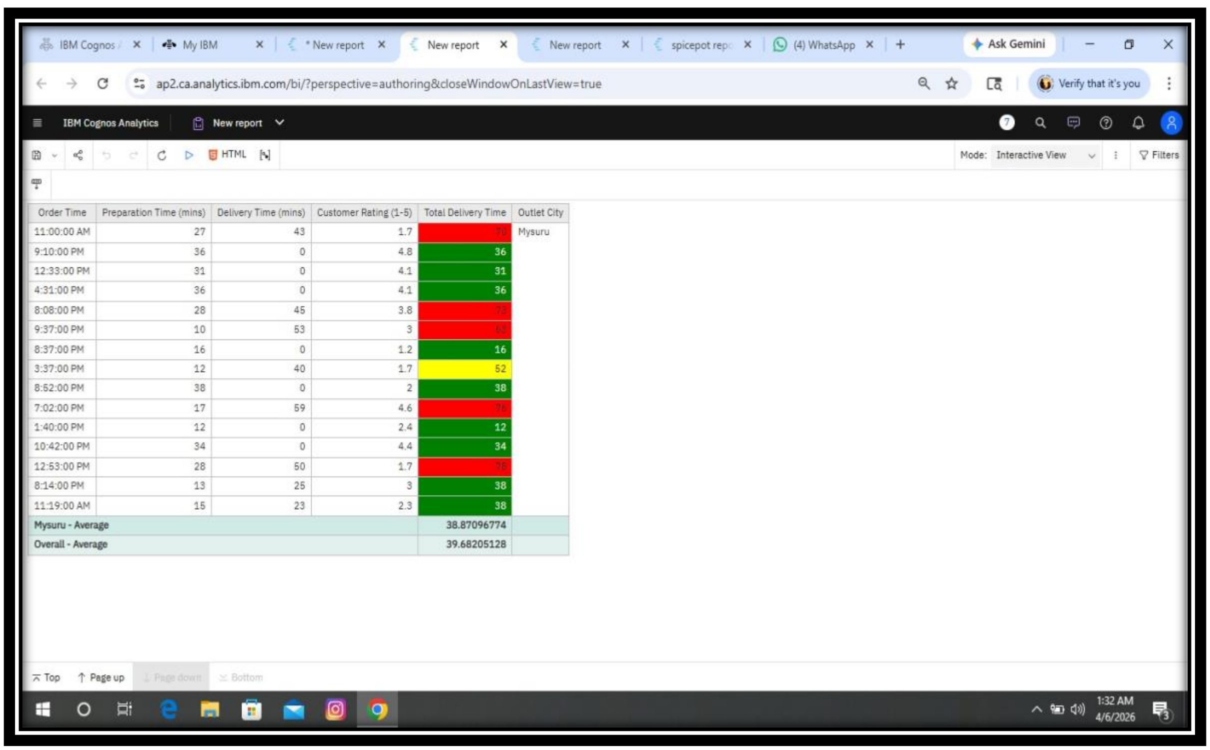
Build a Delivery Performance Report listing all delivery orders with Order Time, Preparation Time, Delivery Time, Total Time, and Customer Rating. Add a calculated field: Total Delivery Time = Preparation + Delivery. Apply Conditional Styling: Total Time > 60 mins in Red, 45–60 in Amber, < 45 in Green. Summarize average delivery time by Outlet City in the Footer.



Order Time	Preparation Time (mins)	Delivery Time (mins)	Customer Rating (1-5)	Total Delivery Time	Outlet City
11:21:00 AM	21	0	3.3	21	Coimbatore
4:34:00 PM	26	0	1.3	26	
Coimbatore - Average				35.42105263	
10:06:00 PM	10	0	2.3	10	Hyderabad
1:52:00 PM	11	0	3.73333333	100	
4:20:00 PM	32	55	1.1	87	
6:16:00 PM	11	38	2.3	49	
11:59:00 AM	38	0	4.5	38	
7:18:00 PM	40	34	1.3	74	
6:29:00 PM	25	0	2.8	25	
10:22:00 PM	11	39	3.4	50	
5:06:00 PM	38	0	3.4	38	
1:20:00 PM	17	0	4.3	17	
12:46:00 PM	16	0	1.3	16	
9:53:00 PM	27	25	2.4	52	
5:46:00 PM	32	55	4.5	87	
6:27:00 PM	10	0	1.5	10	
4:13:00 PM	12	16	3.4	28	
1:36:00 PM	17	19	2.3	36	
2:18:00 PM	39	0	2.9	39	
7:40:00 PM	13	0	2.3	13	



Order Time	Preparation Time (mins)	Delivery Time (mins)	Customer Rating (1-5)	Total Delivery Time	Outlet City
7:55:00 PM	39	0	1.5	39	Coimbatore
12:28:00 PM	24	0	3.6	24	
2:18:00 PM	40	0	3.3	40	
6:05:00 PM	15	20	4.9	35	
3:49:00 PM	24	0	1.5	24	
9:51:00 PM	27	49	4.3	76	
9:19:00 PM	15	0	2.8	15	
11:42:00 AM	44	22	2.3	66	
12:14:00 PM	33	56	2.6	89	
6:04:00 PM	21	0	4.2	21	
4:43:00 PM	45	26	1.8	71	
6:10:00 PM	13	0	2.3	13	
4:58:00 PM	31	0	4.1	31	
4:37:00 PM	10	17	1.4	27	
8:21:00 PM	44	0	4.1	44	
8:32:00 PM	30	0	3.3	30	
1:50:00 PM	23	0	4.6	23	
12:25:00 PM	39	0	2.4	39	
10:18:00 AM	21	0	3.1	21	
2:32:00 PM	26	0	2.3	26	



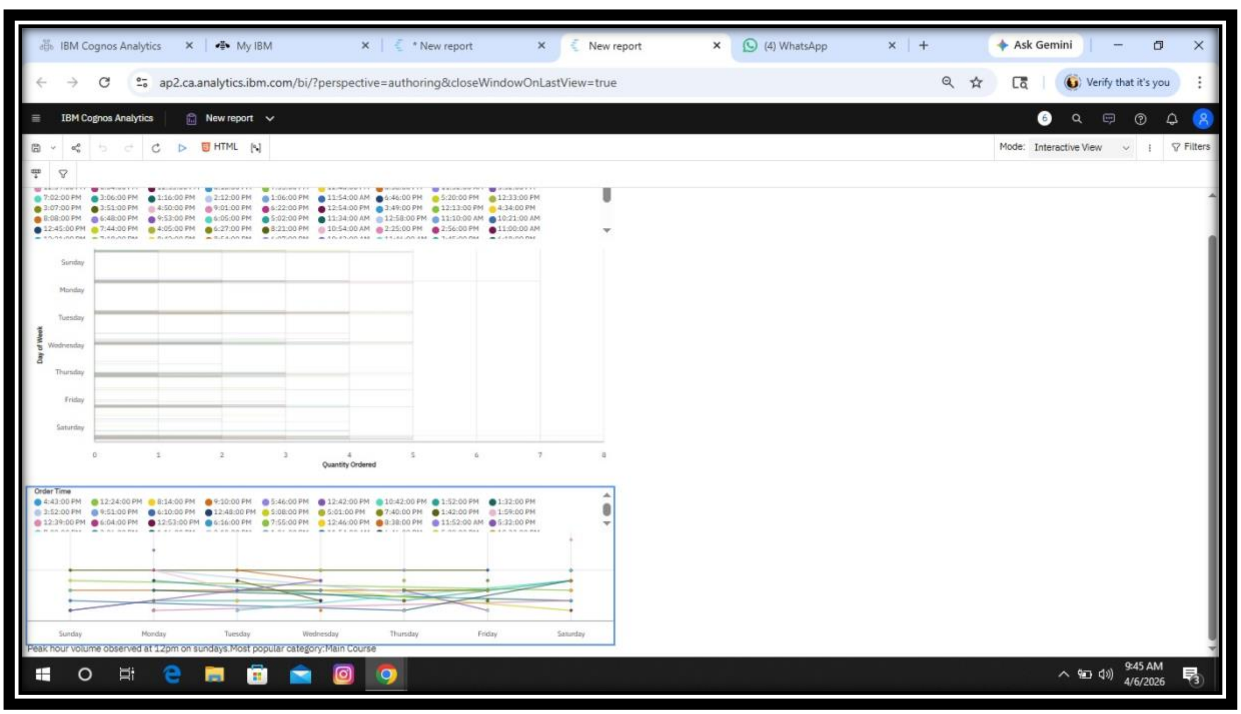
Order Time	Preparation Time (mins)	Delivery Time (mins)	Customer Rating (1-5)	Total Delivery Time	Outlet City
11:00:00 AM	27	43	1.7	70	Mysuru
9:10:00 PM	36	0	4.8	36	
12:33:00 PM	31	0	4.1	31	
4:31:00 PM	36	0	4.1	36	
8:08:00 PM	28	45	3.8	73	
9:37:00 PM	10	53	3	63	
8:37:00 PM	16	0	1.2	16	
3:37:00 PM	12	40	1.7	52	
8:52:00 PM	38	0	2	38	
7:02:00 PM	17	59	4.6	76	
1:40:00 PM	12	0	2.4	12	
10:42:00 PM	34	0	4.4	34	
12:53:00 PM	28	50	1.7	78	
8:14:00 PM	13	25	3	38	
11:19:00 AM	15	23	2.3	38	
Mysuru - Average				38.87096774	
Overall - Average				39.68205128	

Delivery Performance Report(Explanation):

In IBM Cognos Analytics, a Delivery Performance Report is created to analyze the efficiency of order delivery in the SpicePot restaurant chain. The report includes fields such as Order Time, Preparation Time, Delivery Time, Total Time, and Customer Rating. A calculated field called Total Delivery Time is created by adding Preparation Time and Delivery Time. Conditional formatting is applied where Total Time greater than 60 minutes is shown in red, 45–60 minutes in amber, and less than 45 minutes in green. The report helps to identify delivery delays and improve service efficiency. Average delivery time by Outlet City is also summarized in the report footer.

Q3 [REPORTING]

Using Report Visualize Prompt, allow Outlet Managers to select chart type (Bar or Line) to view Hourly Order Volume Trends across different days of the week. Add prompts for Outlet Name and Order Type. Summarize peak hour, busiest day, and most popular category as Text objects. Apply Conditional Styling to highlight hours with Cancelled orders > 10 in Amber.

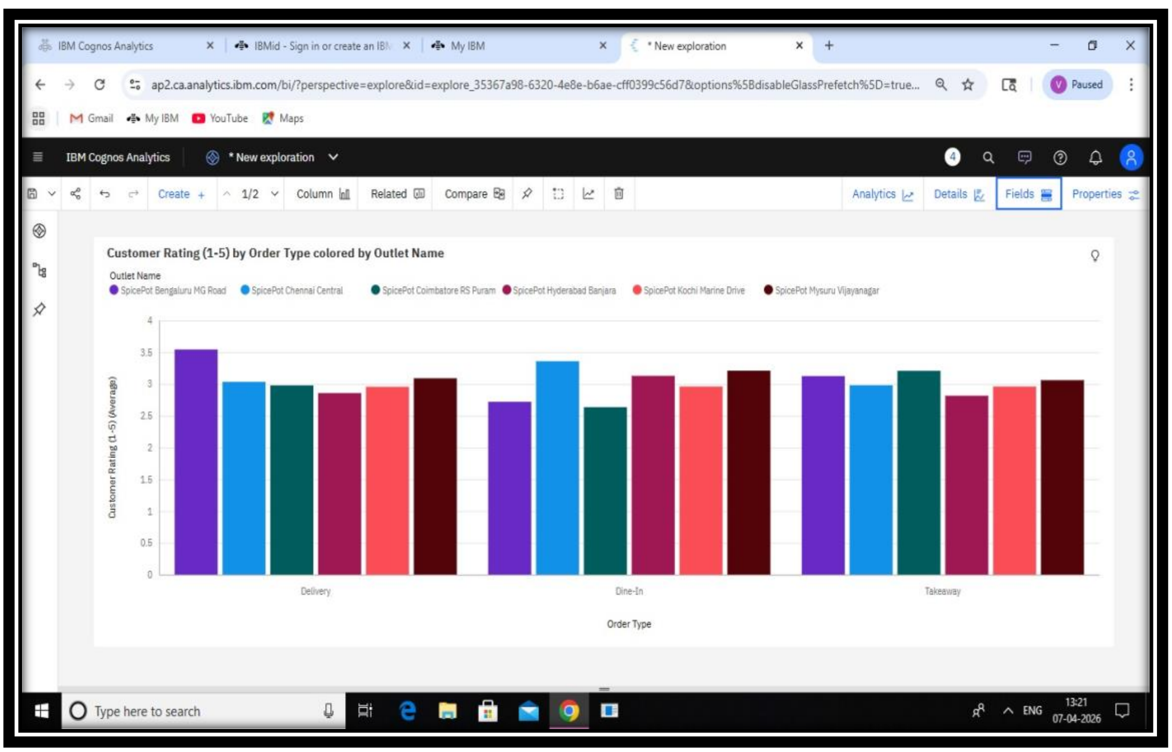


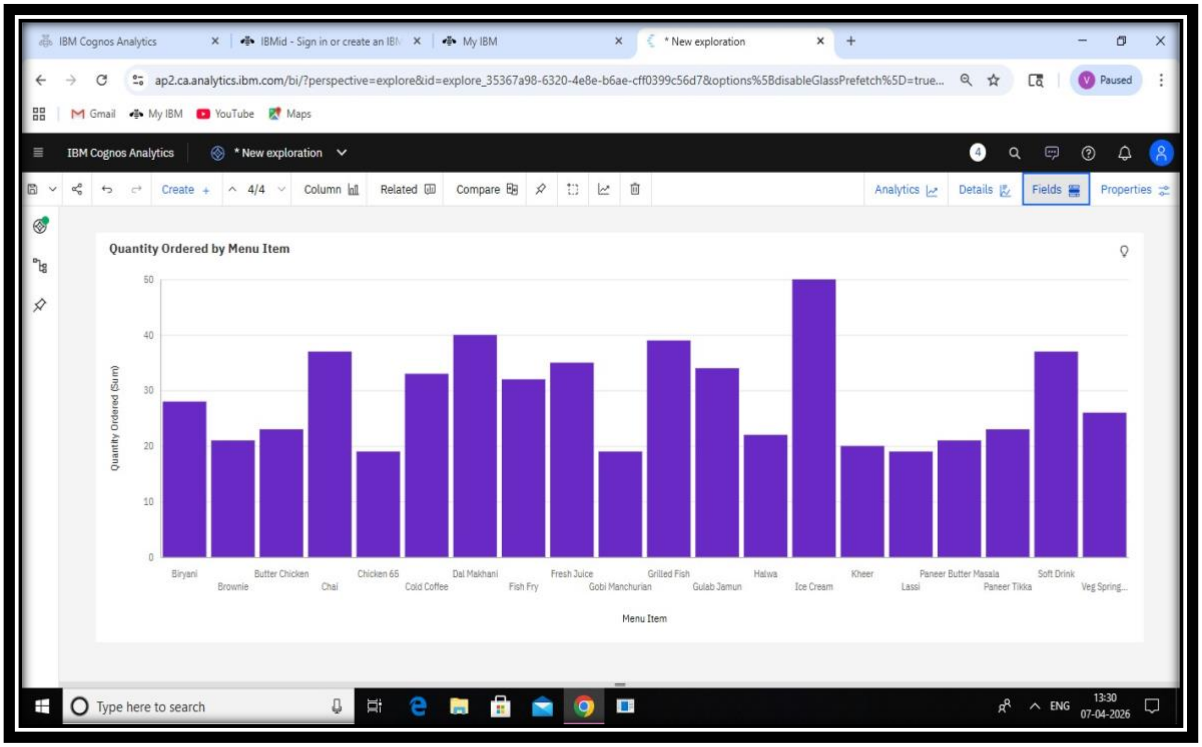
Hourly Order Trend Report(Explanation):

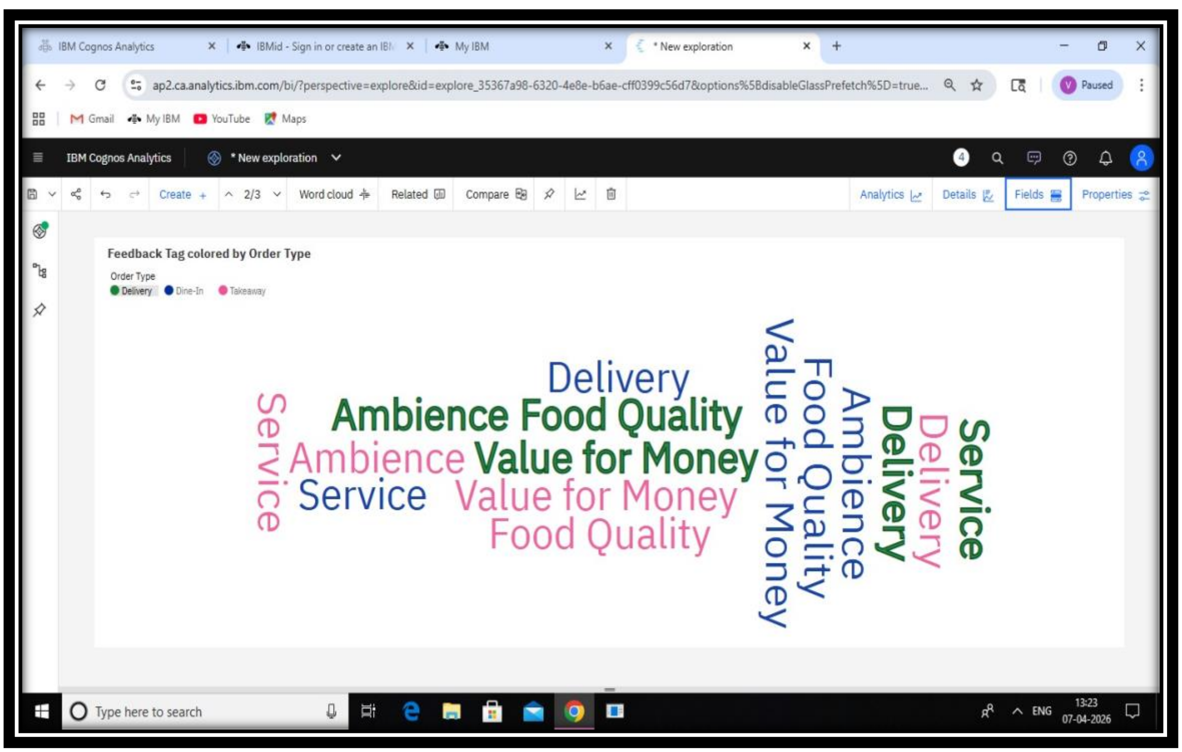
In IBM Cognos Analytics, an Hourly Order Trend Report is created to analyze the number of orders placed at different hours of the day. Managers can select the chart type such as Bar or Line to visualize the hourly order volume trends across different days of the week. Prompts are added for Outlet Name and Order Type to filter the data easily. The report helps identify peak hours, busiest days, and popular order categories. Conditional formatting is applied to highlight hours with cancelled orders greater than 10 in amber color. This analysis helps managers improve service planning and resource allocation.

Q4 [EXPLORATION]

Use Cognos Exploration to discover patterns in the SpicePot dataset. Explore the Relationship between Order Type (Dine-In/Takeaway/Delivery) and Customer Rating. Investigate Which Feedback Tags appear most frequently for Delivery orders and how they correlate with Delivery Time. Identify the top 5 best-performing menu items across all outlets and document Findings.




IBM Cognos Analytics
* New exploration
Quantity Ordered by Menu Item
Quantity Ordered (Sum)
50
40
30
20
10
0
Biryani
Brownie
Butter Chicken
Chai
Chicken 65
Cold Coffee
Dal Makhani
Fish Fry
Fresh Juice
Gobi Manchurian
Grilled Fish
Gulab Jamun
Halwa
Ice Cream
Kheer
Lassi
Paneer Butter Masala
Paneer Tikka
Soft Drink
Veg Spring...
Menu Item

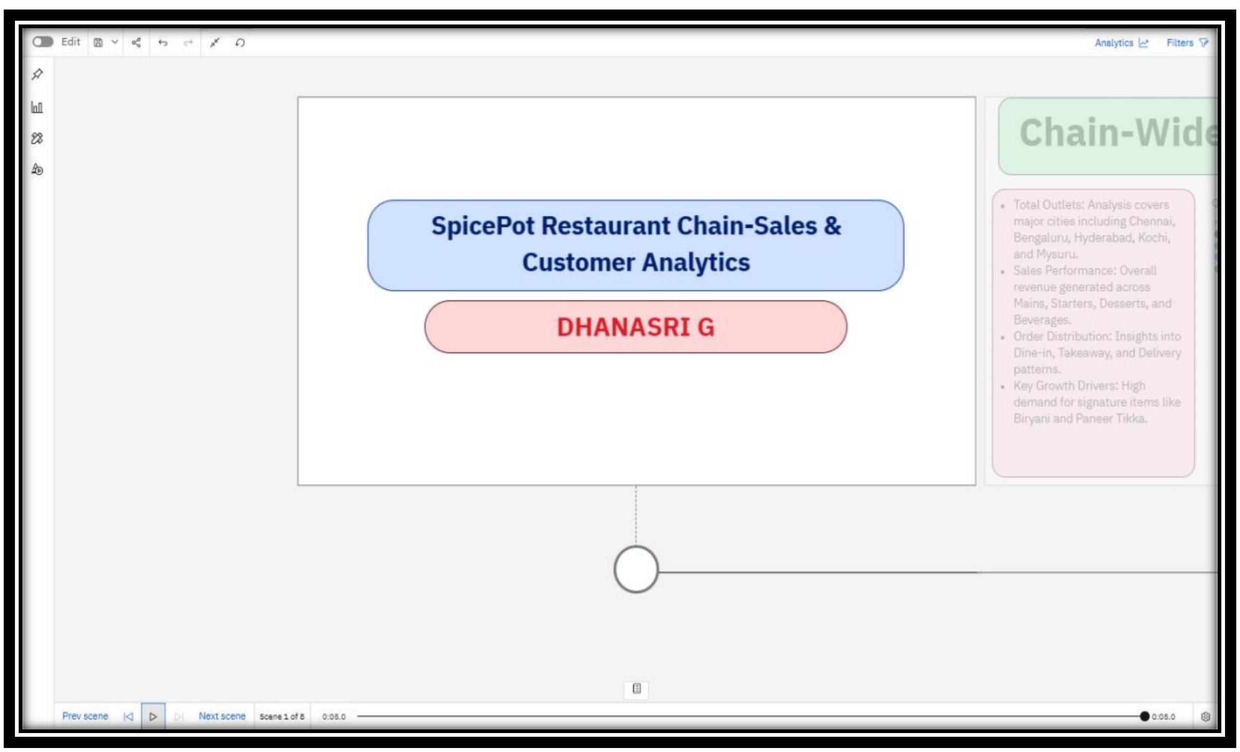

IBM Cognos Analytics
* New exploration
Feedback Tag colored by Order Type
Order Type
Delivery
Dine-In
Takeaway
Delivery
Ambience Food Quality
Service
Ambience
Value for Money
Service
Value for Money
Food Quality
Value for Money
Food Quality
Ambience
Delivery
Delivery
Service

Data Exploration(Explanation):

Using IBM Cognos Analytics Exploration, patterns and relationships in the SpicePot dataset are analyzed. The relationship between Order Type (Dine-In, Takeaway, Delivery) and Customer Rating is examined to understand customer satisfaction. Feedback tags that appear most frequently for delivery orders are identified and their impact on delivery time is analyzed. The exploration also helps to discover trends and hidden insights in the data. Finally, the top 5 best-performing menu items across all outlets are identified based on sales performance and findings are documented.

Q6 [STORY MAKING]

Create a Cognos Story for SpicePot's Monthly Operations Review: Scene 1 —Chain-wide Sales & Order Overview, Scene 2 — Top & Bottom Performing Menu Items, Scene 3 — Delivery Performance & Complaint Analysis, Scene 4 — Customer Satisfaction by Outlet & Order Type, Scene 5 — Peak Hour & Demand Pattern Analysis, Scene 6 — Menu Optimization & Delivery Improvement Recommendations.

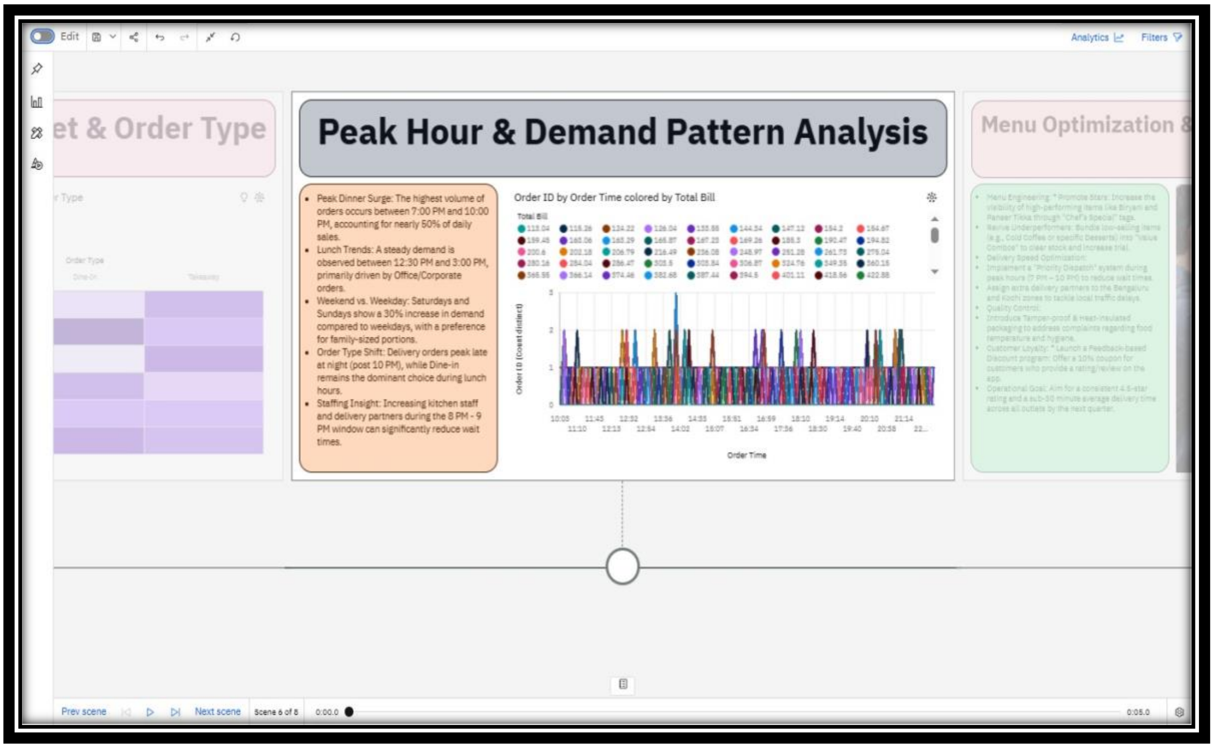



Edit
Analytics
Filters
Chain-Wide Sales & Order Overview
Total Outlets: Analysis covers major cities including Chennai, Bengaluru, Hyderabad, Kochi, and Mysuru.
Sales Performance: Overall revenue generated across Mains, Starters, Desserts, and Beverages.
Order Distribution: Insights into Dine-in, Takeaway, and Delivery patterns.
Key Growth Drivers: High demand for signature items like Biryani and Paneer Tikka.
Quantity Ordered by Menu Item
Menu Item
Chicken 65
Gobi Manchurian
Lassi
Kheer
Brownie
Paneer Butter Masala
Halwa
Paneer Tikka
Butter Chicken
Veg Spring Roll
Biryani
Fish Fry
Cold Coffee
Gulab Jamun
Fresh Juice
Soft Drink
Chai
Grilled Fish
Dal Makhani
Ice Cream
Top & Bottom
Prev scene
Next scene
Scene 2 of 8


Edit
Analytics
Filters
der Overview
Top & Bottom Performing Menu Items
Best Sellers: Biryani and Paneer Tikka are the most frequently ordered items across all outlets.
Revenue Drivers: The Mains and Starters categories contribute the highest percentage to the total sales revenue.
Underperforming Items: Cold Coffee and certain Dessert items show lower sales volume compared to other categories.
Customer Preferences: High demand is observed for signature dishes, especially during weekends and peak dinner hours.
Strategic Recommendation: Introduce promotional offers or "Combo Deals" for bottom-performing items to increase their visibility and sales.
SPICE POT
Delivery Perfo
Prev scene
Next scene
Scene 3 of 8

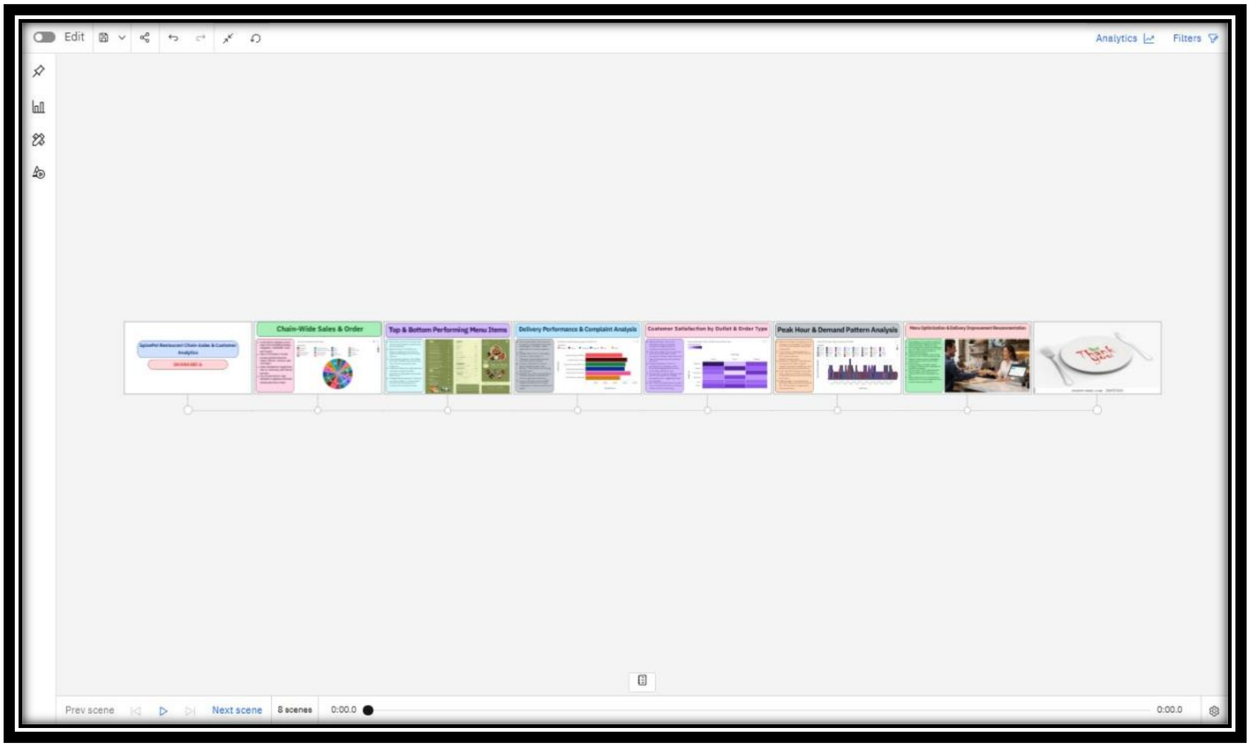

Edit
Analytics
Filters
Menu Items
Delivery Performance & Complaint Analysis
Customer Satis
Order Success Rate: Most orders are successfully "Completed," but a small percentage of "Cancelled" orders needs attention to reduce revenue loss.
Average Delivery Time: Tracking the time from "Order Placed" to "Delivered" to ensure food reaches customers while it's still hot.
Complaint Distribution: Common issues identified include "Late Delivery," "Food Quality," and "Wrong Item Delivered."
Staff Efficiency: Performance analysis of delivery partners and waiters to maintain high service standards.
Customer Satisfaction Score (CSAT): Insights derived from the 1–5 star rating system to improve overall brand loyalty.
Total Bill by Outlet Name colored by Outlet City
Outlet City
Bengaluru
Chennai
Coimbatore
Hyderabad
Kochi
Mysuru
SpicePot Bengaluru MG Road
SpicePot Chennai Central
SpicePot Coimbatore RS Puram
SpicePot Hyderabad Banjara
SpicePot Kochi Marine Drive
SpicePot Mysuru Vijayanagar
Outlet Name
Total Bill (Sum)
Prev scene
Next scene
Scene 4 of 8
0:00.0
0:05.0


Edit
Analytics
Filters
plaint Analysis
Customer Satisfaction by Outlet & Order Type
Peak Hour &
Regional Ratings: Chennai and Hyderabad outlets consistently maintain the highest customer satisfaction scores (4.5+ Stars).
Order Type Insight: Dine-in customers generally provide higher ratings compared to Delivery, likely due to the fresh food experience and restaurant ambiance.
Takeaway Feedback: There is a growing trend in Takeaway orders; ensuring packaging quality is key to maintaining satisfaction here.
Service Gaps: Outlets in Bengaluru and Kochi show a slight dip in ratings during peak hours, suggesting a need for more staff.
Goal: Standardizing the service quality across all SpicePot outlets to achieve a uniform 4.5-star brand rating.
Customer Rating (1-5) by Outlet City and Order Type
Order Type
Delivery
Dine-In
Takeaway
Bengaluru
Chennai
Coimbatore
Hyderabad
Kochi
Mysuru
Outlet City
Prev scene
Next scene
Scene 5 of 8
0:00.0
0:05.0


Edit
Analytics
Filters
et & Order Type
Peak Hour & Demand Pattern Analysis
Menu Optimization &
- Peak Dinner Surge: The highest volume of orders occurs between 7:00 PM and 10:00 PM, accounting for nearly 50% of daily sales.
- Lunch Trends: A steady demand is observed between 12:30 PM and 3:00 PM, primarily driven by Office/Corporate orders.
- Weekend vs. Weekday: Saturdays and Sundays show a 30% increase in demand compared to weekdays, with a preference for family-sized portions.
- Order Type Shift: Delivery orders peak late at night (post 10 PM), while Dine-in remains the dominant choice during lunch hours.
- Staffing Insight: Increasing kitchen staff and delivery partners during the 8 PM - 9 PM window can significantly reduce wait times.
Order ID by Order Time colored by Total Bill
Total Bill
Order ID (Count distinct)
Order Time
Prev scene
Next scene
Scene 6 of 8


Edit
Analytics
Filters
tern Analysis
Menu Optimization & Delivery Improvement Recommentation
- Menu Engineering: * Promote Stars: Increase the visibility of high-performing items like Biryani and Paneer Tikka through "Chef's Special" tags.
- Revive Underperformers: Bundle low-selling items (e.g., Cold Coffee or specific Desserts) into "Value Combos" to clear stock and increase trial.
- Delivery Speed Optimization:
- Implement a "Priority Dispatch" system during peak hours (7 PM – 10 PM) to reduce wait times.
- Assign extra delivery partners to the Bengaluru and Kochi zones to tackle local traffic delays.
- Quality Control:
- Introduce Tamper-proof & Heat-insulated packaging to address complaints regarding food temperature and hygiene.
- Customer Loyalty: * Launch a Feedback-based Discount program: Offer a 10% coupon for customers who provide a rating/review on the app.
- Operational Goal: Aim for a consistent 4.5-star rating and a sub-30 minute average delivery time across all outlets by the next quarter.
Prev scene
Next scene
Scene 7 of 8


Edit
Analytics
Filters
ent Recommentation
shutterstock.com · 70857532
Prev scene
Next scene
Scene 8 of 8
0:00.0
0:05.0


Edit
Analytics
Filters
Prev scene
Next scene
8 scenes
0:00.0
0:00.0

Story Creation(Explanation):

In IBM Cognos Analytics, a story is created to present the analysis results of the SpicePot dataset in a clear and structured way. The story includes slides with charts, dashboards, and key insights about sales, menu performance, and delivery efficiency. Each slide explains the findings using visualizations and short descriptions. It helps managers understand business trends and performance across different outlets. Storytelling makes the data analysis easy to understand and supports better decision-making.

Q5 [DASHBOARD]

Design a SpicePot Operations Dashboard with: (1) KPI tiles — Total Revenue, Average Customer Rating, Order Cancellation Rate, Average Delivery Time, (2) A Bar chart for Revenue by Menu Category, (3) A Line chart for Daily Revenue Trend, (4) A Pie chart for Order Type Distribution (Dine-In/Takeaway/Delivery), (5) A Waiter Performance table by Rating. Add Outlet and Date filters.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>SpicePot Analytics Dashboard</title>
<style>
body{
font-family: Arial, Helvetica, sans-serif;
margin:0;
background:#f4f6f9;
}
/* Login Page */
.login-container{
width:350px;
```

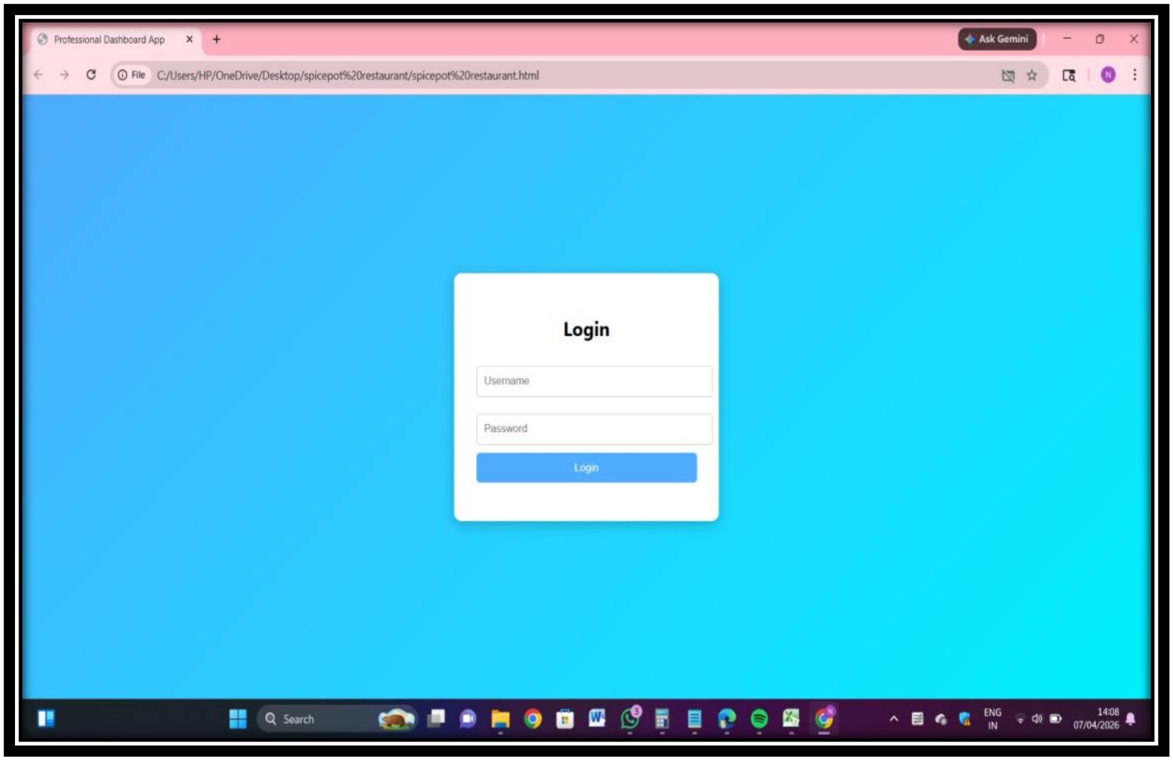
```
margin:120px auto;

background:white;

padding:30px;

border-radius:10px;

box-shadow:0 0 10px rgba(0,0,0,0.2);

text-align:center;

}

.login-container h2{

margin-bottom:20px;

}

input{

width:90%;

padding:10px;

margin:10px 0;

border:1px solid #ccc;

border-radius:5px;

}

button{

width:95%;

padding:10px;

background:#007bff;

border:none;

color:white;

font-size:16px;

border-radius:5px;
```

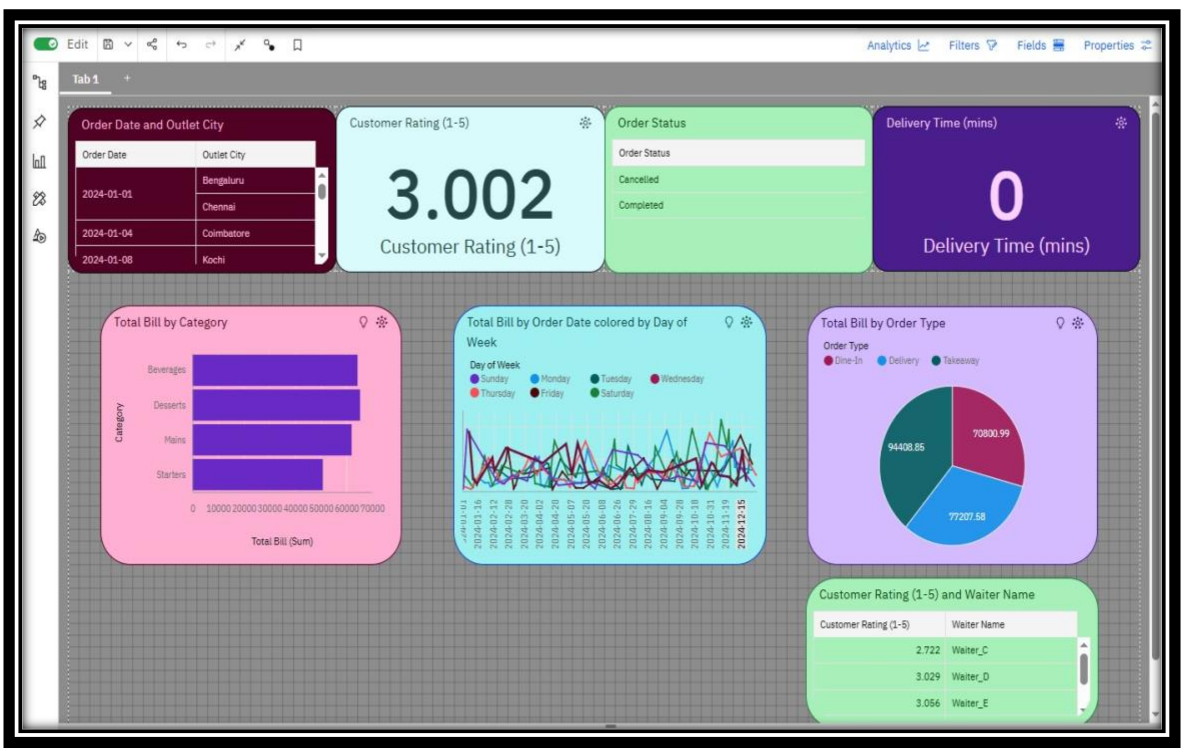
```
cursor:pointer;
}
button:hover{
background:#0056b3;
}
/* Dashboard */
.dashboard{
display:none;
}
.navbar{
background:#222;
color:white;
padding:15px;
display:flex;
justify-content:space-between;
}
.logout{
background:red;
padding:6px 12px;
border:none;
color:white;
cursor:pointer;
border-radius:5px;
}
iframe{
```

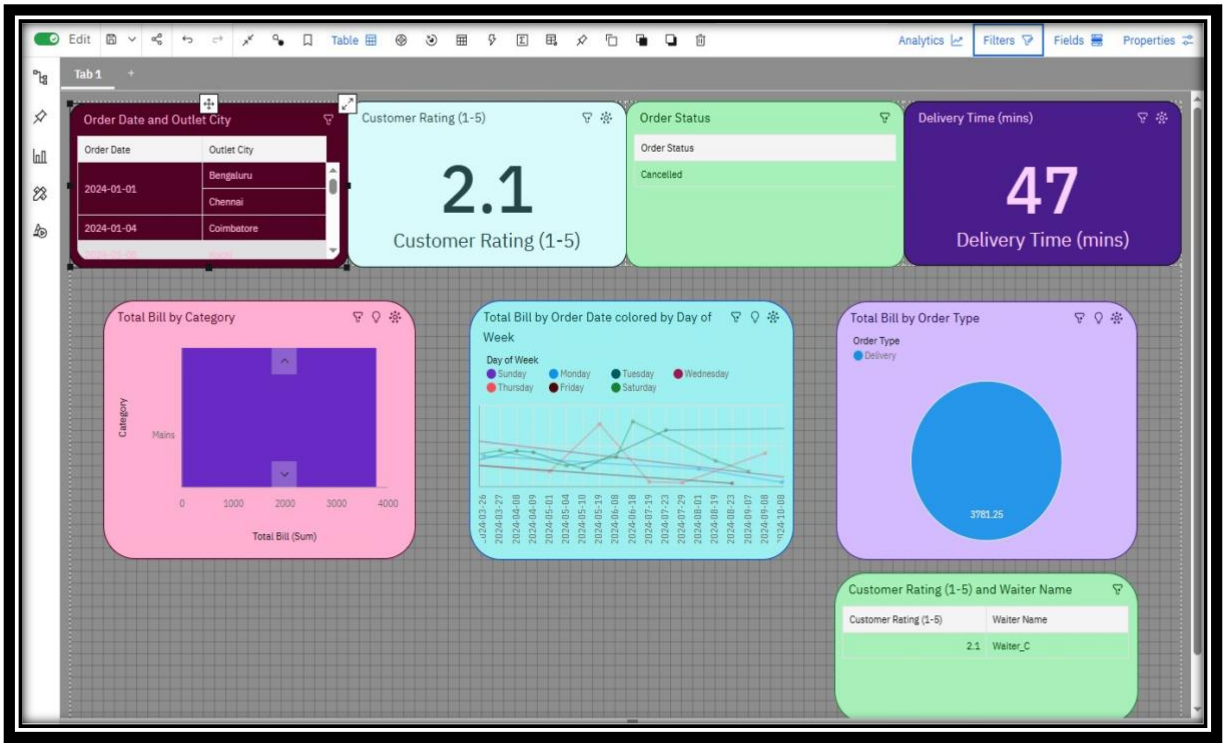
```
width:100%;

height:900px;

border:none;

}

</style>

</head>

<body>

<!-- Login Page -->

<div class="login-container" id="loginPage">

<h2>SpicePot Login</h2>

<input type="text" id="username" placeholder="Enter Username">

<input type="password" id="password" placeholder="Enter Password">

<button onclick="login()">Login</button>

<p id="error" style="color:red;"></p>

</div>

<!-- Dashboard Page -->

<div class="dashboard" id="dashboardPage">

<div class="navbar">

<h3>SpicePot Restaurant Analytics</h3>

<button class="logout" onclick="logout()">Logout</button>

</div>

<iframe
src="https://ap2.ca.analytics.ibm.com/bi/?perspective=dashboard&pathRef=.my_folders
%2FQuestion%2B5&closeWindowOnLastView=true&ui_appbar=false&ui_navbar=false&s
hareMode=embedded&action=view&mode=dashboard&subView=model0000019d6a7fec
6d_00000000&nav_filter=true">
```

```
</iframe>

</div>

<script>

/* Login Function */

function login(){

var user=document.getElementById("username").value;

var pass=document.getElementById("password").value;

/* Simple Login Authentication */

if(user=="admin" && pass=="1234"){

document.getElementById("loginPage").style.display="none";

document.getElementById("dashboardPage").style.display="block";

}

else{

document.getElementById("error").innerHTML="Invalid Username or Password";

}

}

/* Logout */function logout(){

Document.getElementById("dashboardPage").style.display="none";


Document.getElementById("loginPage").style.display="block";

}

</script>

</body>

</html>
```

OUTPUT :


Professional Dashboard App
C:/Users/HP/OneDrive/Desktop/spicepot%20restaurant/spicepot%20restaurant.html
Login
Username
Password
Login


Order Date and Outlet City
Customer Rating (1-5)
3.002
Order Status
Cancelled
Completed
Delivery Time (mins)
0
Total Bill by Category
Total Bill by Order Date colored by Day of Week
Total Bill by Order Type
Customer Rating (1-5) and Waiter Name



Dashboard Creation(Explanation):

In IBM Cognos Analytics, a dashboard is created to visually present important insights from the SpicePot dataset. The dashboard includes charts such as sales by category, top-performing menu items, and hourly order trends. Filters are added for Outlet City and Order Type to allow users to interact with the data. Visual elements like bar charts, pie charts, and line charts are used for better understanding of trends. The dashboard helps managers quickly monitor restaurant performance and make better business decisions.